



Vortex-induced vibration of suspenders in the wake of bridge tower by numerical simulation and wind tunnel test



Yongle Li^{a,*}, Haojun Tang^a, Qiaoman Lin^a, Xinzhong Chen^{a,b}

^a Department of Bridge Engineering, Southwest Jiaotong University, Chengdu 610031, China

^b National Wind Institute, Department of Civil, Environmental and Construction Engineering, Texas Tech University, Lubbock, TX 79409-1023, USA

ARTICLE INFO

Keywords:

Suspension bridge
Suspenders
CFD simulation
Wind tunnel test
Vortex-induced vibration

ABSTRACT

This paper presents a study on the generation mechanism of wind-induced vibration of suspenders of suspension bridges in the wake of bridge towers. The computational fluid dynamics (CFD) simulation is carried out to characterize the wind flow in the wake of bridge towers at different wind directions. The time series of force coefficients of suspenders are determined which clearly show the dominant frequency component associated with the vortex-shedding from the towers. A wind tunnel test program is also implemented to measure wind-induced vibration of suspenders at different wind speeds and directions. Results show that vortex-shedding from bridge towers has a strong influence on the aerodynamic forces of suspenders at a wide range of wind directions. The suspenders vibrate in both longitudinal and transverse directions. The large-amplitude vibration of suspenders is observed when the vortex-shedding frequency from the towers coincides with the frequency of the suspenders.

1. Introduction

Wind-resistant performance is one of the most important concerns in the design of long-span bridges. With the increase of bridge span, wind effects on bridges are increasingly significant. In addition to the static and dynamic buffeting responses and potential flutter instability under strong wind conditions, vibrations of towers and suspenders of suspension bridges caused by vortex-shedding and wake flow need to be carefully investigated and mitigated through mechanical and aerodynamic countermeasures. A number of suspension bridges have been reported to experience wind-induced vibration of long suspenders nearby the tower. As the major connecting component for force transmission, the safety of suspenders is directly related to the safety of the whole bridge structure.

Vortex-induced vibration (VIV) of rectangular and circular cylinder has been investigated extensively in literature (e.g., Okajima, 1982; Knisely, 1990; Kim et al., 2004; Matsumoto et al., 2008; Bearman, 2011; Belloli et al., 2012; Chern et al., 2014; Hua et al., 2015). The aerodynamic interference and wake-induced vibration (WIV) of multiple cylinders have also been addressed (e.g., Zdravkovich and Pridden, 1977; Matsumoto et al., 1990; Brika and Laneville, 1997; Kim et al., 2009a, b; Carmo et al., 2011; Gustavo, 2014; Wang et al., 2014; Lee et al., 2015). In this paper, the generation mechanism of wind-induced vibration of suspenders of suspension bridges in the wake of bridge

towers is examined. The computational fluid dynamics (CFD) simulation is carried out to characterize the wind flow in the wake of bridge towers and the forces on the suspenders. The time series of force coefficients of suspenders are determined which clearly show the frequency component associated with the vortex-shedding from the towers. A wind tunnel test program is also implemented to measure wind-induced vibration of suspenders at different wind speeds and directions. Results show that vortex-shedding from bridge towers is responsible to the large-amplitude vibration of suspenders in both longitudinal and transverse directions.

2. CFD simulation

2.1. Analysis model of CFD

For long-span suspension bridges, two-column tower is the most common form. The elevation diagram of a long-span suspension bridge with two-column towers is shown in Fig. 1. Six suspenders which are the most close to bridge tower are considered. Considering the fact that the inclination of tower column is small and the size changes very little along the height, this study use two dimensional (2D) approach to investigate the general aerodynamic characteristics of the suspenders. The geometric configurations are taken at the half-height of the tower as shown in Fig. 2.

* Corresponding author.

E-mail address: lele@swjtu.edu.cn (Y. Li).

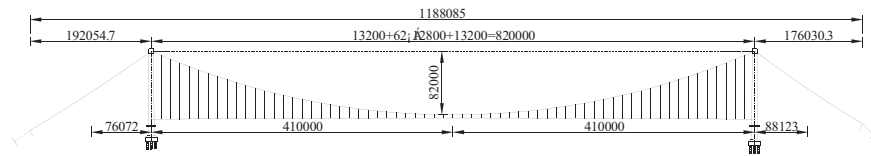


Fig. 1. Elevation graph of structure (unit: mm).

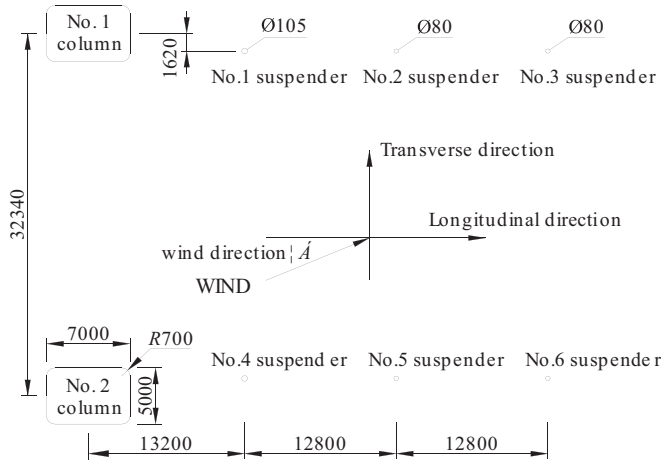


Fig. 2. Schematic diagram of cross-section size (unit: mm).

Table 1

Aerodynamic coefficients of No.1 suspender.

Items	Re	Mean C_H	St
Present results	10^5	1.155	0.191
Schlichting and Gersten (2003)	10^5	1.250	0.190
Bi et al. (2012)	10^5	1.250	0.200

Table 2

Aerodynamic coefficients of No.1 tower column.

Items	Side ratio B/D	Re	St	Mean C_H	RMS C_V
Present results	1.40	6.6×10^4	0.114	1.736	0.696
Sakamoto et al. (1989)	1.50	5.5×10^4	0.113	–	0.700
Norberg (1993)	1.50	1.3×10^4	0.117	1.700	–
Shimada and Ishihara (2002)	1.50	2.2×10^4	0.108	1.600	0.500

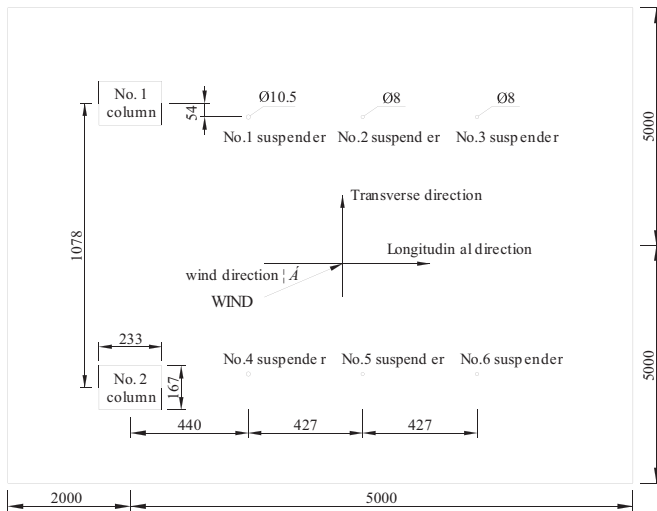


Fig. 3. Cross-section size and calculating region (unit: mm).

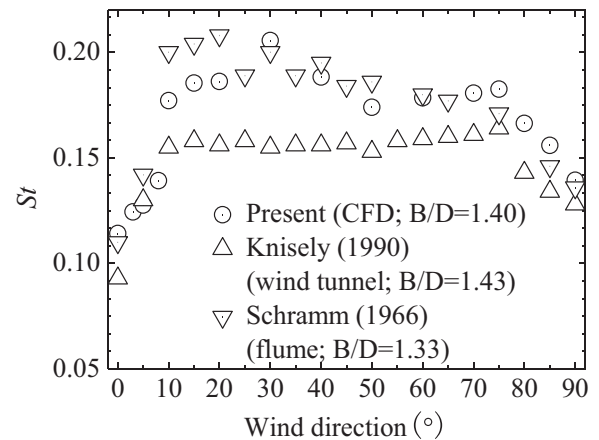


Fig. 5. Strouhal number of rectangular section at different wind directions (Schramm, 1996).

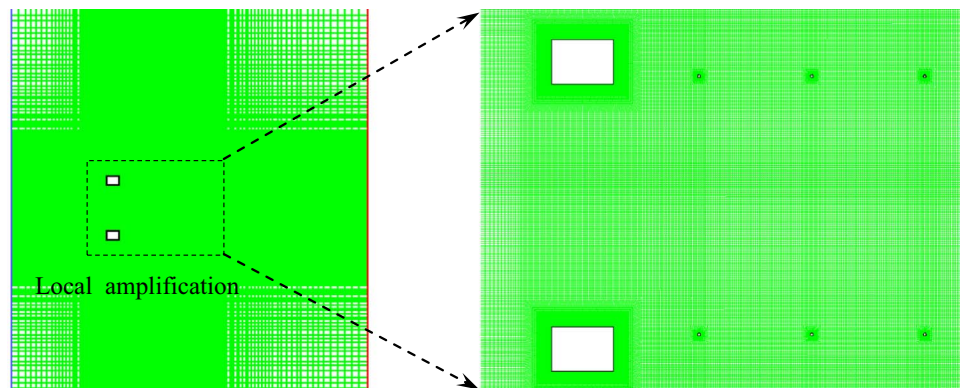


Fig. 4. Local schematic diagram of grid partition.

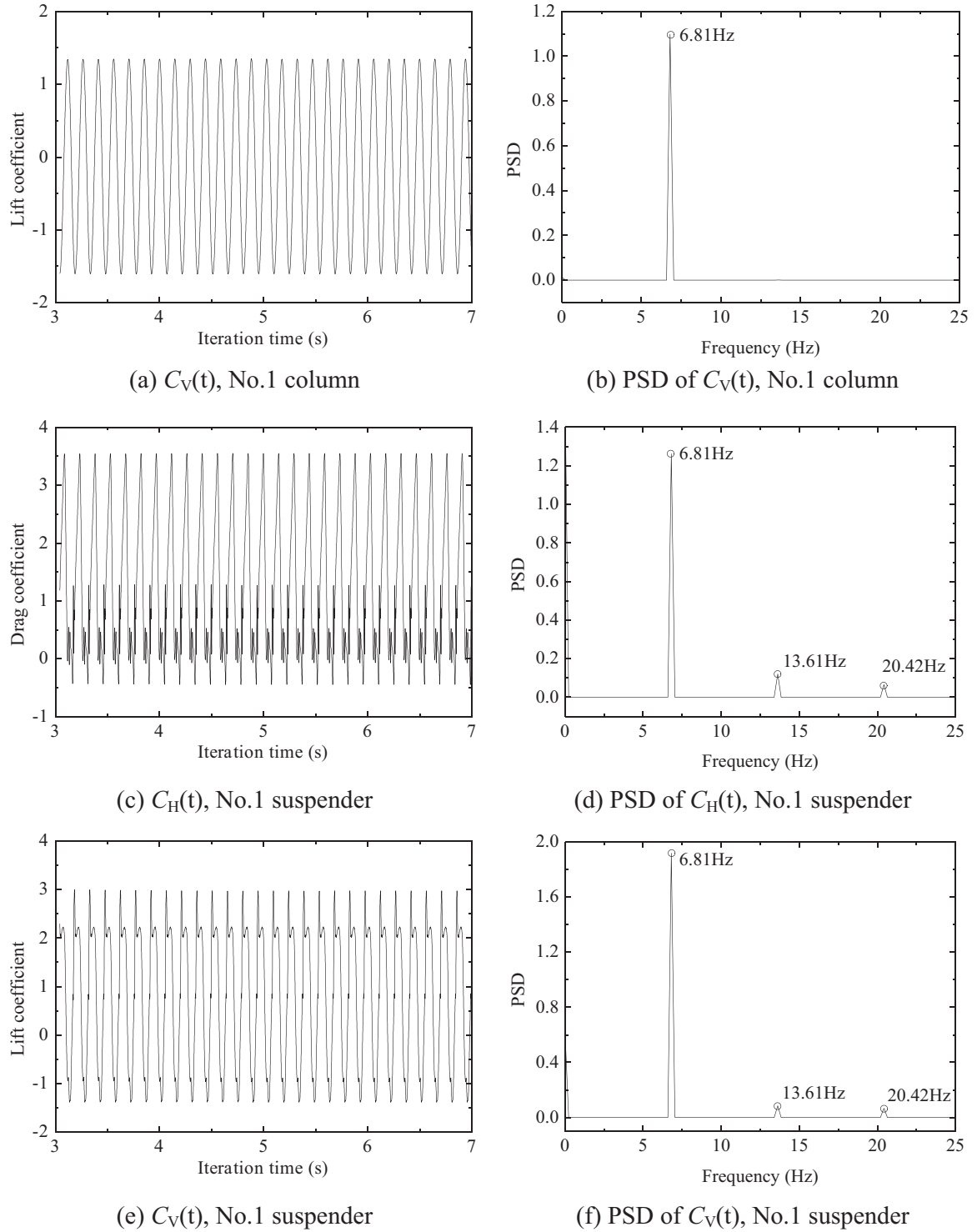


Fig. 6. Time series and PSDs of drag and lift coefficients at 0° wind direction.

In the wind tunnel testing program to be discussed later, the scale ratios of tower columns and suspenders are set as 1/30 and 1/10 to satisfy the requirement of blockage ratio and to consider the maximum wind speed of the XNJD-1 wind tunnel in Southwest Jiaotong University of China. Considering the fact that the aerodynamic interference of suspenders on the bridge tower can be neglected, a larger size ratio of suspenders is used to improve the accuracy of measurement. The tower cross sections are simplified as rectangular sections without consideration of the round corners to avoid the potential Reynolds number effects. Same sizes of towers and suspenders are used

in the CFD simulation for the convenience of comparison. The Reynolds number of tower column is about 10^5 and the Reynolds numbers of suspenders are between 10^3 and 10^4 at the calculation wind speed.

The size of computational domain is shown in Fig. 3. The boundary of windward side is set as velocity-inlet and the boundary of leeward side is set as pressure-outlet. The boundaries of upper side and lower side are set as velocity-inlet or pressure-outlet which will depend on the condition of inflow, and the boundaries of tower columns and suspenders are set as smooth wall. The computational mesh, discrete

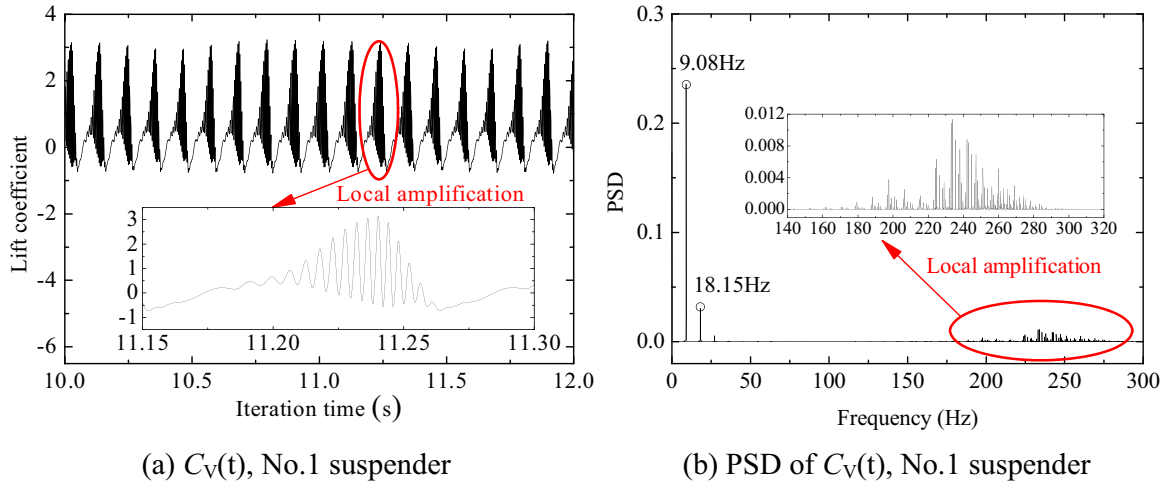


Fig. 7. Time series and PSDs of lift coefficient at 10° wind direction.

Table 3

Dominant frequencies of lift coefficients of suspenders at different wind directions.

Wind direction	Column (Hz):		Suspender (Hz):					
	No.1	No.2	No.1	No.2	No.3	No.4	No.5	No.6
0°	6.81	6.81	6.81	6.81	6.81	6.81	6.81	6.81
10°	9.08	8.24	9.08	9.08	9.08	8.24	8.24	8.24
20°	7.84	8.67	7.84	7.84	7.84	8.67	8.67	8.67
30°	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83
45°	6.50	6.50	6.50	6.50	6.50	6.50	6.50	–
60°	5.47	5.47	5.47	5.47	5.47	5.47	–	–
75°	6.62	6.62	6.62	6.62	6.62	6.62	–	–
90°	5.92	5.92	5.92	–	–	–	–	–

Note: only the dominant frequencies for the maximum amplitude are listed in Table 3, and “–” represents the dominant frequencies for the maximum amplitude are larger than 100 Hz.

tized by quadrilateral structured grid, is shown in Fig. 4 and the element size progressively increased from the sides of tower columns and suspenders to the computational boundary. The $k-\omega$ SST turbulence model is adopted and the dimensionless time-step is set equal to 10^{-4} . The discretized problem has been numerically solved by adopting a SIMPLE pressure-velocity coupling algorithm. A second order scheme is selected for pressure, and a second order upwind scheme is selected for momentum, turbulent kinetic energy, and specific dissipation rate. The unsteady flow is simulated by CFD software FLUENT.

2.2. Discussion of the results

For the cross sections of suspender and column, the drag and lift force coefficients are defined as

$$C_H(\alpha) = F_H(\alpha) / (\frac{1}{2} \rho U^2 D) \quad (1)$$

$$C_V(\alpha) = F_V(\alpha) / (\frac{1}{2} \rho U^2 B) \quad (2)$$

where $F_H(\alpha)$ and $F_V(\alpha)$ are the drag and lift forces respectively, and the direction along wind is positive; D and B are the widths of cross-section perpendicular and parallel to the wind, respectively; ρ is the air density; U is the mean wind velocity; and α is the angle of attack.

The Strouhal number St is defined as

$$St = fD/U \quad (3)$$

where f is the vortex-shedding frequency.

At first, the unsteady flow around single suspender and that around single column are simulated. Taking the No.1 suspender and the No.1 column as examples, the results at 0° angle of attack are shown in Tables 1, 2. The results are consistent to those reported in literature, which demonstrates the adequacy of the simulation scheme. According to Eq. (3), the wind speed for vortex-induced vibration can be calculated, and the value of the No.1 suspender is 0.59 m/s for its first natural frequency (1.073 Hz). In this situation, if there are no interferences caused by other components, the vortex-induced vibration of suspender will not occur due to the low vortex lock-in wind speed.

The wake flow characteristics of single column at different wind

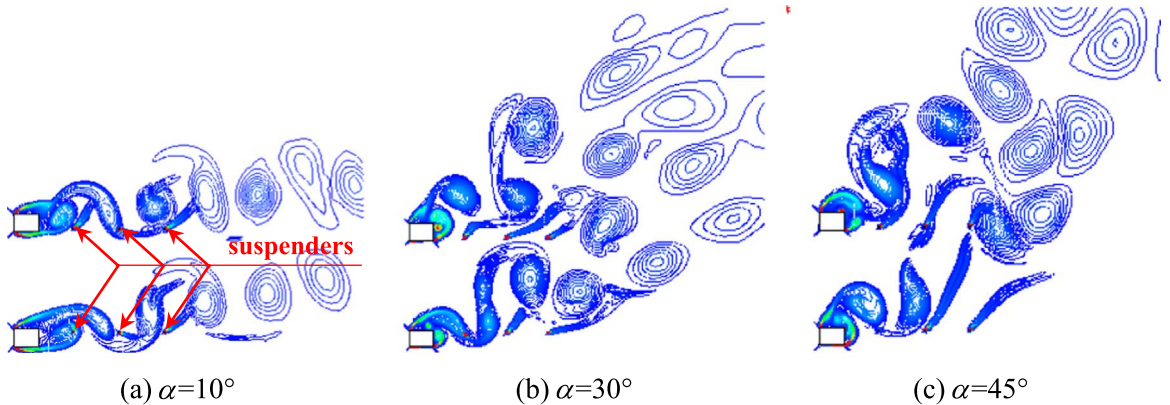


Fig. 8. The instantaneous vorticity at different wind directions.

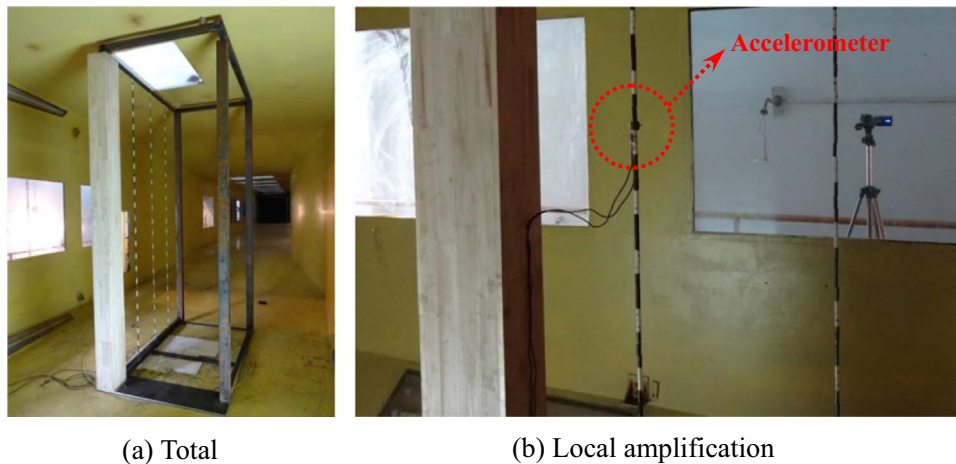


Fig. 9. Wind tunnel test model.

Table 4
Parameters of column and suspenders in wind tunnel test.

Member	Size of cross-section (mm)	Length (m)	Mass per unit length ($\text{g}\cdot\text{m}^{-1}$)	Damping ratio (%)	First order natural frequency (Hz)
No.1 column	167×233	2.86	—	—	—
No.1 suspender	Φ10.5	2.78	403	0.30	2.799
No.2 suspender	Φ8	2.78	259	0.91	3.664
No.3 suspender	Φ8	2.78	259	0.87	4.123
No.5 suspender	Φ8	2.78	259	0.84	3.373

Note: the tower column is rigidly fixed to the two ends.

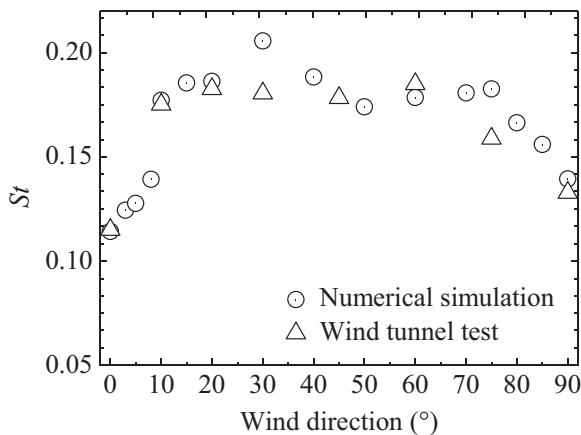


Fig. 10. Strouhal number of tower column by wind tunnel test.

directions ranging from 0° to 90° with wind speed of 10 m/s are then simulated. The change of St with wind direction is shown in Fig. 5. When the wind direction is less than 10° , the value of St increases with the increase in wind direction. When the wind direction is between 10° and 75° , the value of St is relatively stable. When the wind direction is larger than 75° , the value of St decreases with the increase in wind direction.

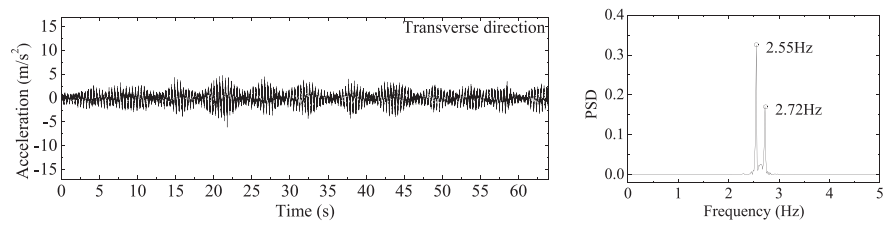
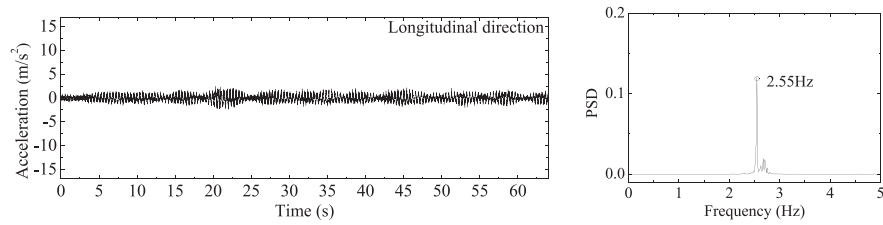
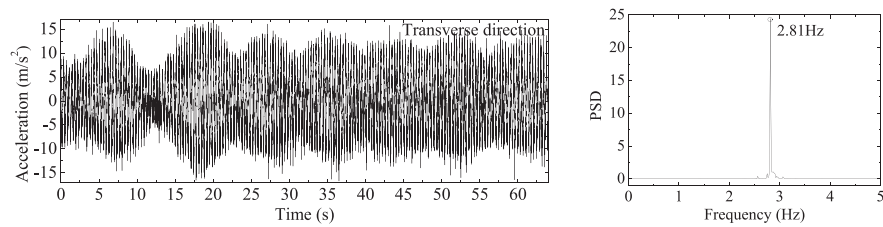
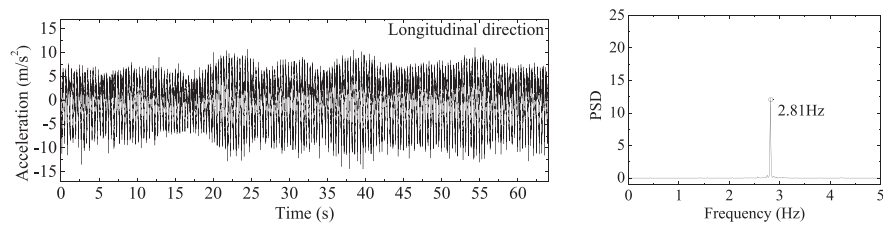
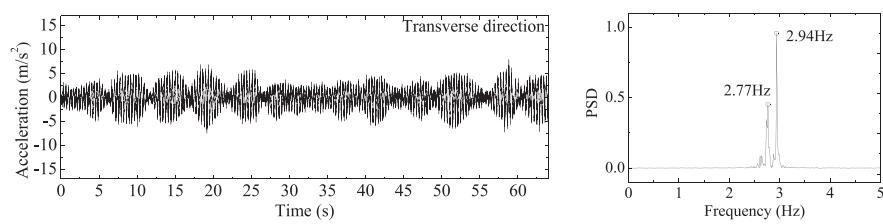
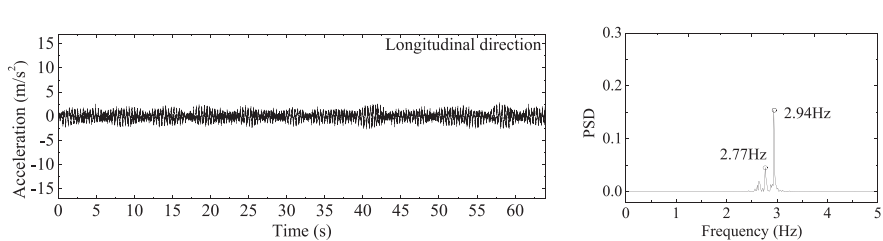
Subsequently, all the six suspenders and the two columns have been taken into consideration. Fig. 6 shows the time series and PSDs of drag and lift coefficients at 0° wind direction. The dominant frequencies of lift and drag coefficients of the suspender are identical to the vortex-shedding frequency of the two tower columns. Fig. 7 displays the time series and PSDs of lift coefficient at 10° wind direction. At 10° degrees, the time series of the lift coefficients of the suspenders include low- and high-frequency components. The low-frequency component is caused by vortex-shedding of tower column. The high-frequency component is contributed by a wide range of frequencies from 180 to 280 Hz for the No.1 suspender. The corresponding reduced frequencies fD/U are

between 0.189 and 0.294 which are close to the Strouhal number St of the suspender. Therefore, the high-frequency component is considered to be related to the vortex-shedding of suspender itself. The dominant frequencies of lift coefficients of suspenders and the tower columns at different wind directions are summarized in Table 3. The instantaneous vorticity at different wind directions is shown in Fig. 8. When the wind direction is between 0° and 30° , the dominant frequencies of lift coefficients of all six suspenders are identical to the vortex-shedding frequencies of tower columns. The influence is stronger for the suspenders closer to the tower. The proportion of the high-frequency component increases with the increase in wind direction, which indicates that the effect of wake flow on suspenders is gradually weakened. When the wind direction is increased to 45° , the dominant frequency of the lift coefficient of No.6 suspender is no longer identical to the vortex-shedding frequency of the tower column. With the further increase in wind direction, these suspenders will gradually leave the wake region.

3. Wind tunnel test

3.1. Experimental setup

As mentioned previously, the scale ratios of tower column and suspenders are set as 1/30 and 1/10 to satisfy the requirement of blockage ratio. The tower column is made of wood and rigidly fixed to the two ends. The suspender consists of a steel wire with plastic and steel cylinders attached. The length of each cylinder is 2.5 cm. The frequency of suspender is determined by adjusting the tension of the steel wire. As shown in Fig. 9, a steel frame is designed to fix the tower column and the suspenders. The parameters of suspenders are summarized in Table 4. A cobra probe located at longitudinal direction is used to measure the fluctuating wind speed in the wake of the tower. The suspender vibrations in two directions are monitored through two accelerometers installed at 1/3 height of the suspender. Compared to the location of the middle height, the accelerometers can capture the

(a) Transverse direction for $U=3.65\text{m/s}$ (b) Longitudinal direction for $U=3.65\text{m/s}$ (c) Transverse direction for $U=4.03\text{m/s}$ (d) Longitudinal direction for $U=4.03\text{m/s}$ (e) Transverse direction for $U=4.21\text{m/s}$ (f) Longitudinal direction for $U=4.21\text{m/s}$ **Fig. 11.** Acceleration time series of No.1 suspender at different wind velocities.

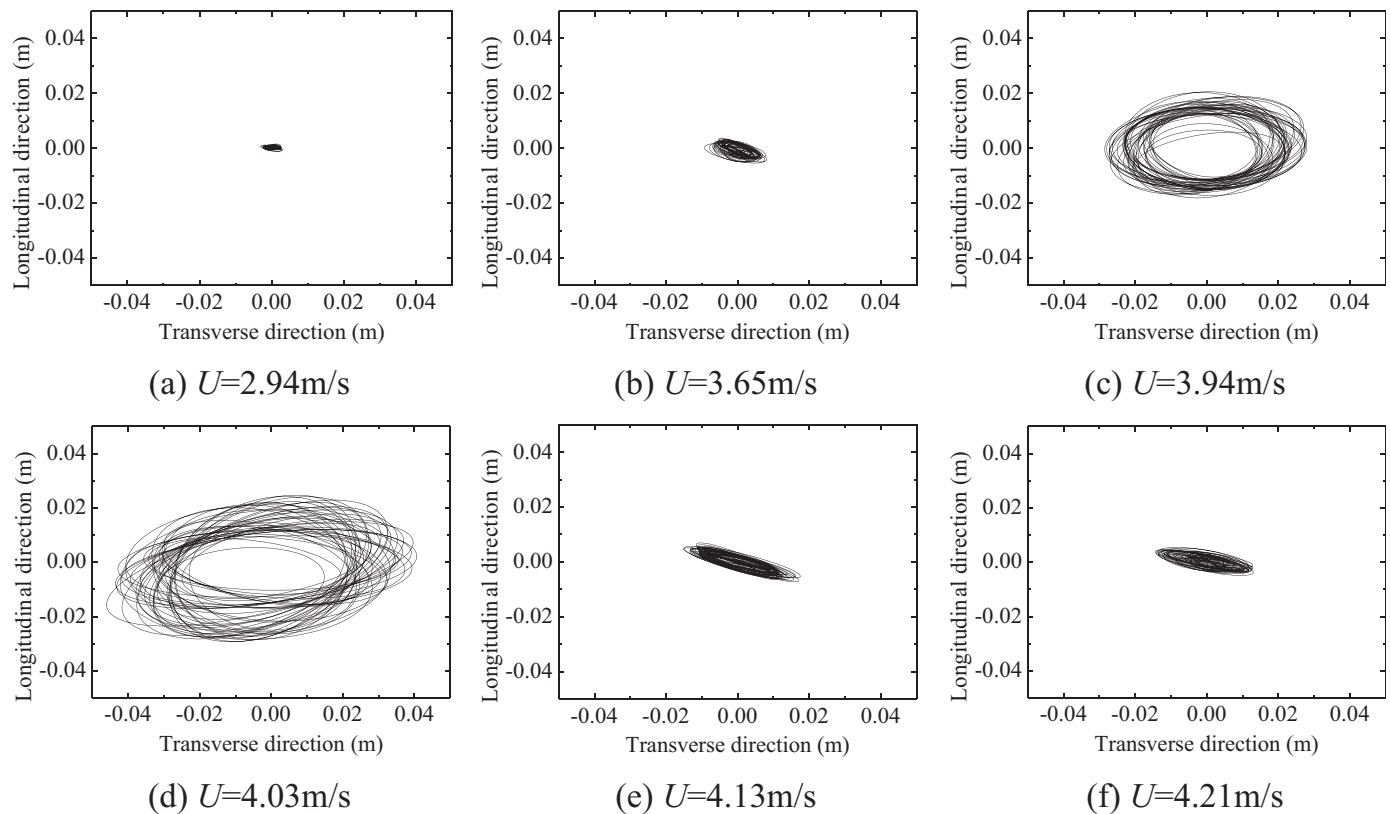


Fig. 12. Trajectories of No.1 suspender at different wind velocities.

vibrations of both first and second modes.

3.2. Discussion of the results

Firstly, the wake flow characteristics of tower column are analyzed at different wind speeds from 4 to 8.5 m/s with 0° wind direction. The measuring point is located at the position of No.2 suspender. It is confirmed that St is fixed and equal to 0.115 which is the same with the result of CFD.

Secondly, the wake flow characteristics of tower column at different wind directions are analyzed. The measuring point is located at the position after the tower column with the distance between $2B$ and $3.5B$, where B is the width of tower column parallel to the wind direction. The wind direction ranges from 0° to 90° , and the wind speed is between 8.40 and 8.55 m/s. The St calculated by wind tunnel test and numerical simulation are shown in Fig. 10, which are quite agreeable.

The accelerations of No.1 suspender in both transverse and longitudinal directions at different wind speeds with 0° wind direction are shown in Fig. 11, in terms of time series and corresponding power spectra. The acceleration in transverse direction is larger than that in the longitudinal direction, but their dominant frequencies are close to each other. At lower wind speed, there are two dominant frequencies. The smaller one is vortex-shedding frequency with larger amplitude. Another is natural frequency of the suspender. With the increase in wind speed, the vortex-shedding frequency approaches to the natural frequency of suspender, so the amplitude of acceleration increases. When the vortex-shedding frequency is equal to the natural frequency of the suspender, the occurrence of resonance is observed, so we see only one frequency and the amplitude of acceleration reaches the maximum. At higher wind speed, we see the two frequencies again, but the vortex-shedding frequency is higher than the natural frequency.

The time series of displacement are obtained through integrations of time series of acceleration. A bandpass filter is firstly used to the original time series of acceleration, preserving the main signal of the

two frequencies. It is followed by double integrations. Finally, the trend noise of the time series of displacement which is produced in the process of integral is removed by a conic fitting. Taking the No.1 suspender as an example with 0° wind direction, the trajectories of suspender at the $1/3$ height are shown in Fig. 12.

The amplitude of suspender increases with the increase in wind speed. When the wind speed is equal to 4.03 m/s, i.e., the reduced wind speed, defined by $U/(f \cdot B)$ where B is the width of tower column parallel to the wind, is 6.18 m/s, the amplitude reaches its maximum, and the amplitude of single peak in transverse direction is about four times as big as the diameter of suspender. Then, the amplitude of suspender decreases rapidly with the increase in wind speed.

The RMS displacements at different reduced wind speeds of No.1-3 suspenders at 0° wind direction and No.5 suspender at 30° wind direction are shown in Fig. 13. For the four suspenders, the resonance caused by the vortex-shedding from the tower column occurs almost at the same reduced wind speed $U/(f \cdot B)$, where the RMS displacements reach the maximum in longitudinal and transverse directions simultaneously. With the decrease in distance between tower and suspender, the maximum RMS displacements increase, because the aerodynamic interference is stronger. With the further increase in wind speed, the second modal vibration of suspender can be observed when the vortex-shedding frequency of tower column is close to the second modal frequency of suspender.

The maximum RMS displacements for resonance of No.1 and No.2 suspenders at different wind directions are shown in Fig. 14. With the increase in wind direction, the maximum RMS displacements show a decreasing trend, but there is a significantly decrease of the value at 10° wind direction.

4. Discussion of countermeasures

Most tower columns of long-span bridges now have been designed with corner modification. For the suspension bridge referred in this

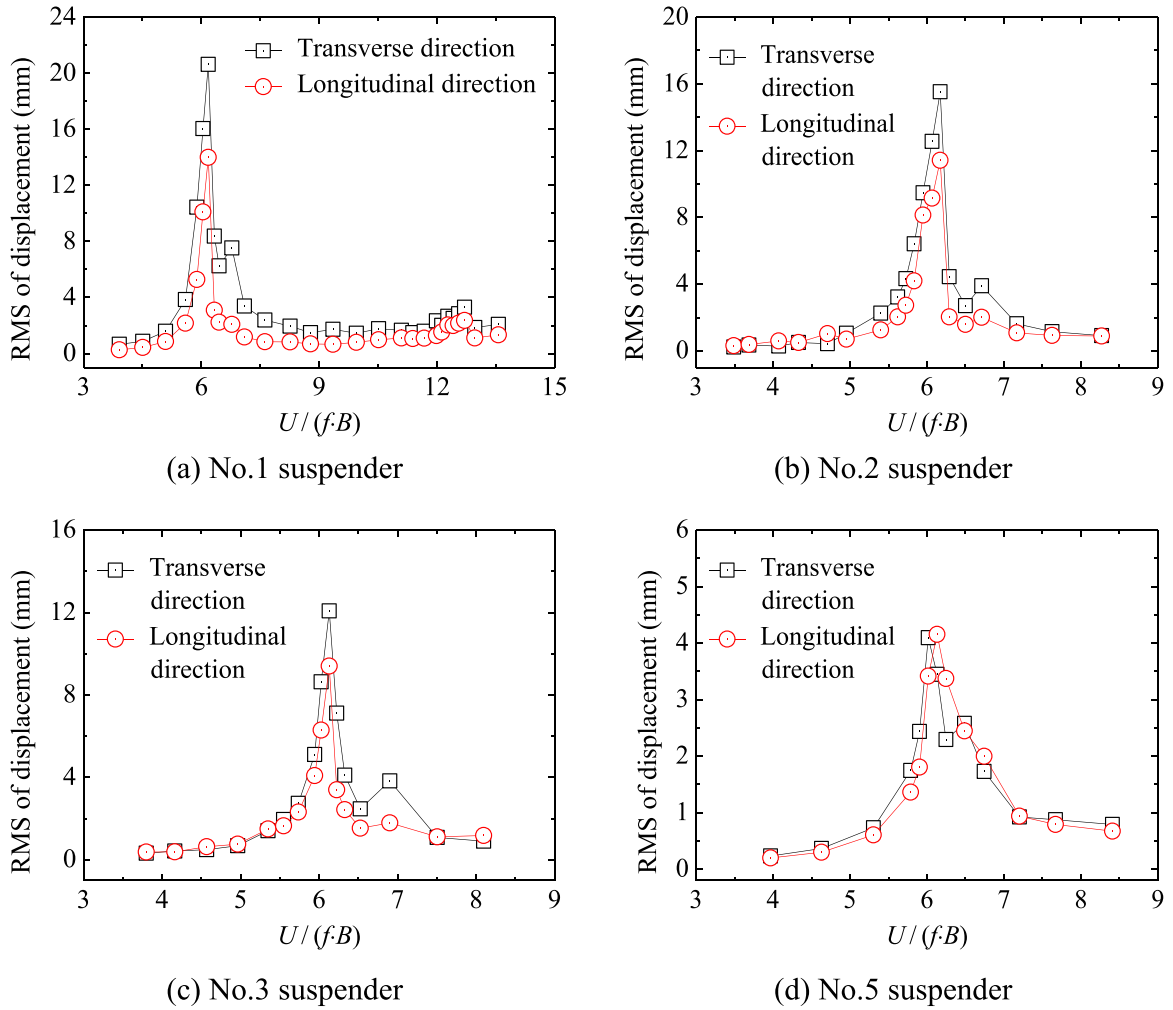


Fig. 13. RMS displacements at different reduced wind speeds.

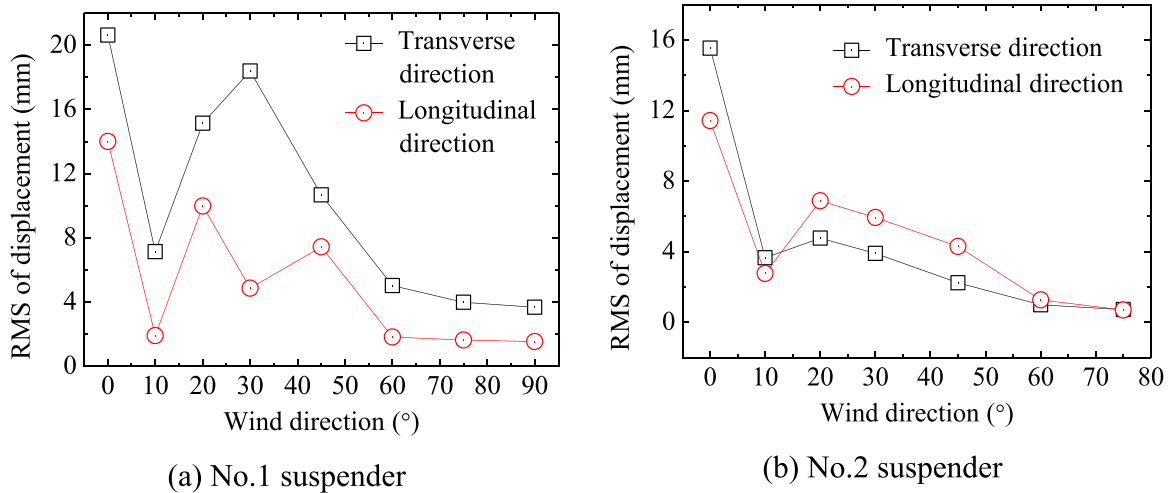


Fig. 14. The maximum RMS displacements at different wind directions.

paper, circular corner-cut with a radius of 0.7 m is adopted, as shown in Fig. 2. The aerodynamic characteristics of the six suspenders in the wake of actual bridge tower are further analyzed by CFD simulation, and the scale ratios of tower columns and suspenders are set as 1/1. The vortex-induced vibration of suspenders is significant at lower wind directions as well. When the wind direction is increased to 70°, only the dominant frequency of the lift coefficient of No.1 suspender is identical

to the vortex-shedding frequency from tower columns. With the further increase in wind direction, all these suspenders leave the wake region. Compared with the bridge tower without circular corner-cut, the influence scope of the bridge tower with circular corner-cut is smaller. Taking 90° wind direction with wind speed of 10 m/s as an example, the instantaneous vorticity of the two tower columns when the lift force of upstream column, i.e., the No.2 column, reaches its maximum and

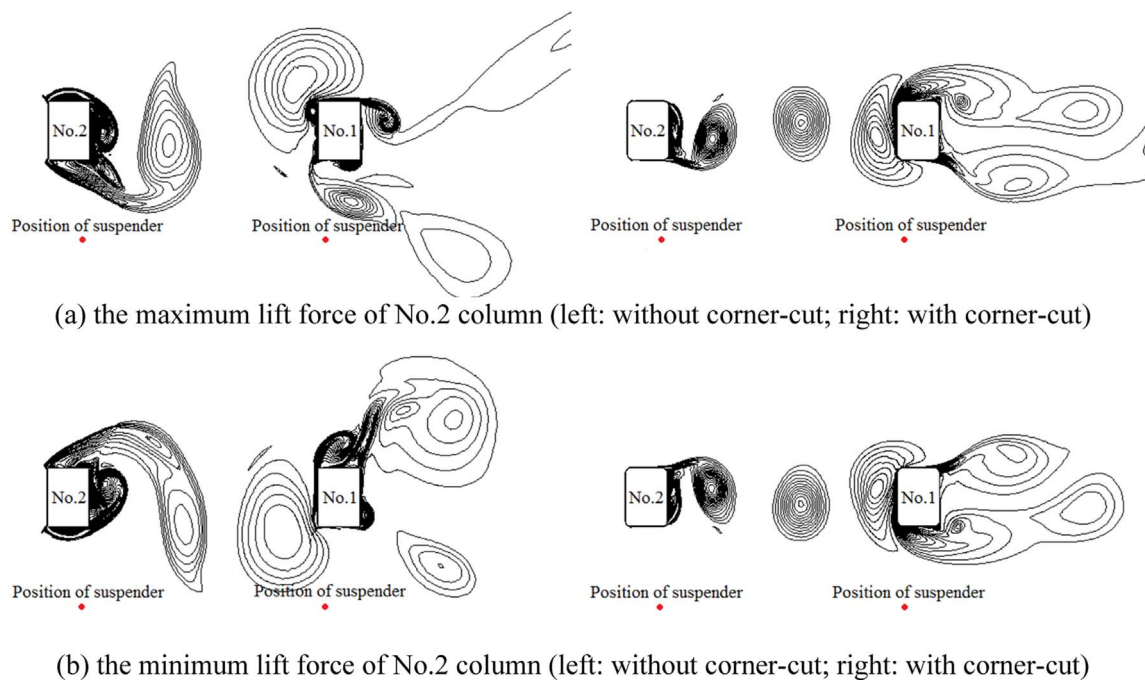


Fig. 15. The instantaneous vorticity at different times (scale ratio=1/1).

minimum are shown in Fig. 15. It can be seen that the reason is the movement of separation point of inflow toward downstream, so the width of wake flow becomes narrow. The existence of circular corner-cut decreases the RMS aerodynamic coefficients of tower columns, which is also favorable to the stability of suspenders. On the other hand, however, the vortex-shedding frequency from tower columns with circular corner-cut increases obviously, leading to the resonance of suspenders occurs at lower wind speeds. Results show that the six suspenders have the possibility of the first modal vibration when wind speed is between 22 m/s and 32 m/s.

To improve the critical wind speed of the resonance of suspenders, it is an essential and effective countermeasure by optimizing the shape of bridge towers and increasing the difference between the vortex-shedding frequency from tower columns and the frequency of suspenders. Moreover, the tower column with variable cross-section, which makes the frequency of the vortex acting on suspenders change along the height, may be favorable to reduction of the resonance of suspenders.

For the suspender itself, as the aerodynamic characteristics are controlled by the wake flow of bridge towers to a great extent, aerodynamic optimization has little effect on improvement of the stability. Structural optimization which increases the natural frequencies and makes the resonance of suspenders occur at higher wind speeds, and mechanical optimization which increases the damping ratio and weakens the resonance of suspenders are both effective countermeasures.

5. Conclusions

The generation mechanism and characteristics of wind-induced vibration of long suspenders nearby the tower in the wake of bridge tower were investigated through CFD simulation and wind tunnel test. It was concluded that the vortex-shedding from bridge towers has a strong influence on the aerodynamic forces of suspenders at a wide range of wind directions. The suspenders vibrate in both longitudinal and transverse directions. The large-amplitude vibration of suspenders was observed when the vortex-shedding frequency from the towers coincides with the frequency of the suspenders. The maximum RMS

displacement of suspender decreases with the increase in distance between tower and suspender, and with the increase in wind direction.

Acknowledgments

The authors are grateful for the financial supports from the National Science Fund for Distinguished Young Scholars (51525804), the Construction Technology Project of China Transport Ministry (2014318800240), the Sichuan Province Youth Science and Technology Innovation Team (2015TD0004) and the National Key Technology R & D Program (2012BAG05B02).

References

- Bearman, P.W., 2011. Circular cylinder wakes and vortex-induced vibrations. *J. Fluids Struct.* 27, 648–658.
- Belloli, M., Giappino, S., Muggiasca, S., Zasso, A., 2012. Force and wake analysis on a single circular cylinder subjected to vortex induced vibrations at high mass ratio and high Reynolds number. *J. Wind Eng. Ind. Aerodyn.* 103, 96–106.
- Bi, J.H., Yu, H.J., Ren, H.P., 2012. Two dimensional numerical simulation of flow over a static square cylinder and a static circular cylinder. *J. China Three Gorges Univ. (Nat. Sci.)* 34 (1), 41–45, (in Chinese).
- Brika, D., Laneville, A., 1997. Wake interference between two circular cylinders. *J. Wind Eng. Ind. Aerodyn.* 72, 61–70.
- Carmo, B.S., Sherwin, S.J., Bearman, P.W., et al., 2011. Flow-induced vibration of a circular cylinder subjected to wake interference at low Reynolds number. *J. Fluids Struct.* 27 (4), 503–522.
- Chern, M.J., Kuan, Y.H., Nugroho, G., Lu, G.T., Horng, T.L., 2014. Direct-forcing immersed boundary modeling of vortex-induced vibration of a circular cylinder. *J. Wind Eng. Ind. Aerodyn.* 134, 109–121.
- Gustavo, R.S.A., 2014. Wake-induced vibration of tandem cylinders of different diameters. *J. Fluids Struct.* 50, 329–339.
- Hua, X.G., Chen, Z.Q., Chen, W., Niu, H.W., Huang, Z.W., 2015. Investigation on the effect of vibration frequency on vortex-induced vibrations by section model tests. *Wind Struct.* 20 (2), 349–361.
- Kim, D.H., Yang, K.S., Senda, M., 2004. Large-eddy simulation of turbulent flow past a square cylinder confined in a channel. *Comput. Fluids* 33 (1), 81–96.
- Kim, S., Alam, M.M., Sakamoto, H., Zhou, Y., 2009a. Flow-induced vibrations of two circular cylinders in tandem arrangement. Part 1: characteristics of vibration. *J. Wind Eng. Ind. Aerodyn.* 97, 304–311.
- Kim, S., Alam, M.M., Sakamoto, H., Zhou, Y., 2009b. Flow-induced vibrations of two circular cylinders in tandem arrangement. Part 2: suppression of vibrations. *J. Wind Eng. Ind. Aerodyn.* 97, 312–319.
- Knisely, C.W., 1990. Strouhal numbers of rectangular cylinders at incidence: a review and new data. *J. Fluids Struct.* 4, 371–393.

- Lee, S.J., Kim, J.J., Yeom, E., 2015. Vortex-induced reconfiguration of a tandem arrangement of flexible cylinders. *Wind Struct.* 21 (1), 25–40.
- Matsumoto, M., Shiraishi, N., Shirato, H., 1990. Aerodynamic instabilities of twin circular cylinders. *J. Wind Eng. Ind. Aerodyn.* 33, 91–100.
- Matsumoto, M., Yagi, T., Tamaki, H., Tsubota, T., 2008. Vortex-induced vibration and its effect on torsional flutter instability in the case of $B/D=4$ rectangular cylinder. *J. Wind Eng. Ind. Aerodyn.* 96, 971–983.
- Norberg, C., 1993. Flow around rectangular cylinders: pressure forces and wake frequencies. *J. Wind Eng. Ind. Aerodyn.* 49 (1–3), 187–196.
- Okajima, A., 1982. Strouhal numbers of rectangular cylinders. *J. Fluid Mech.* 123 (1), 379–398.
- Sakamoto, H., Haniu, H., Kobayashi, Y., 1989. Fluctuating force acting on rectangular cylinders in uniform flow. *Trans. Jpn. Soc. Mech. Eng.* 516, 2310–2317.
- Schlichting, H., Gersten, K., 2003. *Boundary-Layer Theory* 8th edition. Springer, New York, USA.
- Schramm, W. (1966). “Wirbelfrequenzmessungen an umströmten Bauteilen”, *fl-Mitteilungen*(Institut für Leichtbau und Ökonomische Verwendung von Werkstoffen Mitteilungen), 5, pp. 308–318.
- Shimada, K., Ishihara, T., 2002. Application of a modified k-e model to the prediction of aerodynamic characteristics of rectangular cross-section cylinders. *J. Fluids Struct.* 16 (4), 465–485.
- Wang, H.K., Yang, W.Y., Nguyen, K.D., Yu, G.L., 2014. Wake-induced vibrations of an elastically mounted cylinder located downstream of a stationary larger cylinder at low Reynolds numbers. *J. Fluids Struct.* 50, 479–496.
- Zdravkovich, M.M., Pridden, D.L., 1977. Interference between two circular cylinders; Series of unexpected discontinuities. *J. Ind. Aerodyn.* 2, 255–270.